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Title: Comparisons between low-intensity resistance training with blood flow restriction and high-intensity resistance training on quadriceps muscle mass and strength in elderly.

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Brief running head: Lower-limb resistance training and blood flow restriction in elderly.

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ABSTRACT

High-intensity resistance training (HRT) has been recommended to offset age-related loss in muscle strength and mass. However, part of the elderly population is often unable to exercise at high-intensities. Alternatively, low-intensity resistance training with blood flow restriction (LRT-BFR) has emerged. The purpose of the present study was to compare the effects of LRT-BFR and HRT on quadriceps muscle strength and mass in elderly. Twenty-three elderly individuals, 14 men and nine women (age: 64.04 ± 3.81 years; weight: 72.55 ± 16.52 Kg; height: 163 ± 11cm), undertook 12 weeks of training. Subjects were ranked according to their pre-training quadriceps cross-sectional area (CSA) values and then randomly allocated into one of the following groups: (1) control group (CG); (2) HRT: 4 X 10 repetitions 70-80% 1-RM; (3) LRT-BFR: 4 sets (1 x 30 and 3 x 15 repetitions) 20-30% 1-RM. The occlusion pressure was set at 50% of maximum tibial arterial pressure and sustained during the whole training session. Leg-press 1-RM and quadriceps CSA were evaluated at pre- and post-training. A mixed-model analysis was performed and the significance level was set at $P \leq 0.05$. Both training regimes were effective in increasing pre- to post-training leg-press 1-RM (HRT: ~54 %, $P < 0.001$; LRT-BFR: ~17 %, $P = 0.067$) and quadriceps CSA (HRT: 7.9 %, $P < 0.001$; LRT-BFR: 6.6 %, $P < 0.001$), however, HRT seems to induce greater strength gains. In summary, LRT-BFR constitutes an important surrogate approach to HRT as an effective training method to induce gains in muscle strength and mass in elderly.

KEY WORDS: Vascular occlusion ; KAATSU, muscle hypertrophy; aging.

INTRODUCTION

Aging is characterized by a number of structural and functional changes in the organism that lead to a progressive loss of muscle strength and mass (7-9, 18). In this regard, physical exercise, namely, resistance training (RT), has been widely recommended to offset some of the age-related impairments in functionality and alterations in body composition (4, 12, 13, 24).

Current recommendations advocate that RT should be performed at least twice a week with moderate to vigorous intensity in order to maintain or enhance muscle strength and mass (2, 3). Despite the previously demonstrated effectiveness of high-intensity RT (HRT) (4, 12, 13, 24, 29) , the heavy load implicated in this form of training may not be suitable to the entire elderly population, including the frail elderly, novice RT practitioners, and older individuals with joint and/or cardiorespiratory impairments (19, 21, 25).

An alternative approach to HRT is the combination of low-intensity RT (*e.g.*, 20-30% one repetition maximum [1RM] with partial blood flow restriction [LRT-BFR]). LRT-BFR has been alleged to induce similar gains in muscle mass and strength as compared to conventional HRT in different populations (*e.g.*, athletes and health young adults) (1, 14, 17, 27, 28). The clinical application of this intervention constitutes an interesting approach to RT in elderly population, as older individuals are often unable to exercise at high intensities.

However, data regarding the effects of LRT-BFR in older individuals are still scarce (14, 31). To the best of our knowledge, only one study has addressed the effects of this training method on lower-limb muscle strength in older individuals (14). The authors reported similar gains in lower-limb 1RM in a direct comparison between LRT-BFR and HRT. On the other hand, no study has provided a comparative assessment of the two methods in respect of its effects on muscle mass in elderly population. The sole study that has investigated the effects of LRT-BFR on muscle mass in older individuals (31) did not have an HRT group, warranting further investigations.

Thus, the aim of the present study was to compare the effects of 12 weeks of either LRT-BFR or HRT on quadriceps muscle strength and mass in elderly individuals.

METHODS

Experimental Approach to the Problem

We used a repeated measures design to assess the effectiveness of LRT-BFR in increasing the muscle strength and mass in older individuals. Subjects were ranked in quartiles according to their initial quadriceps CSA and then randomly allocated into one of the following groups: control (CG - n = 7; 66.0 ± 5.0 yrs; 69.3 ± 16.1 kg; 1.61 ± 0.1 m), high-intensity resistance training (HRT - n = 8; 62.0 ± 3.0 yrs; 68.7 ± 15.3 kg; 1.60 ± 0.1 m) and low-intensity resistance training with blood flow restriction group (LRT-BFR - n = 8; 65.0 ± 2.0 yrs; 79.3 ± 17.9 kg; 1.70 ± 0.1 m). Two weeks prior to the commencement of the training program (PRE), quadriceps CSA [assessed by magnetic resonance imaging (MRI)] and muscle strength (assessed by the 1-RM test) were evaluated. Importantly, the subjects performed two familiarization sessions to get acquainted with the leg-press 1-RM test procedures before testing. The 1-RM test was performed 48 hours after the last familiarization session. Afterwards, subjects underwent 12 weeks of training two times per week. The control group maintained daily live activities. The leg press 1-RM was reassessed at the 6th week to adjust the training loads. The quadriceps CSA and leg-press 1-RM were also assessed at the end of the intervention (POST: five days after the completion of the last training session).

Subjects

Twenty three healthy older individuals (14 men and nine women) volunteered to participate in this study. The inclusion criteria were not to have cardiac disease, arterial hypertension, diabetes mellitus or any musculoskeletal conditions in the lower extremities that precluded the participation in the training protocols and tests proposed and not to have participated in a RT program for at least six months prior to the study. All of the participants provided their informed consent and the experimental protocol was approved by the University’s ethics committee. The study was conducted in conformity with the policy statement regarding the use of human subjects by the Declaration of Helsinki.

1-RM Test

Leg-press strength was assessed using the 1-RM test on the leg press machine (45° leg press G3-PL70 MATRIX, Indaiatuba, São Paulo, Brazil) following the Brown and Weir recommendations (5). Briefly, the protocol consisted of a five-minute general warm-up on an ergometric bicycle at 60 rpm and 25 watts. This was followed by a specific warm-up consisted of one set of ten repetitions at 50% of the estimated 1-RM, followed by one set of three repetitions at 70% of the estimated 1-RM with one-minute rest between sets. After a three-minute rest period, the subjects ha up to five attempts to achieve their 1RM. A three-minute rest interval was respected between attempts and the higher load achieved (fully eccentric-concentric movement with 90° range of motion) was considered as the 1-RM.

Quadriceps CSA

Quadriceps CSA was obtained through MRI (Signa LX 9.1; GE Healthcare, Milwaukee, WI). Participants were positioned lying in a supine position with the knees extended and legs straight. An initial reference image was obtained to determine the perpendicular distance from the greater trochanter of the femur to the inferior border of the lateral epicondyle of the femur, which was defined as the segment length. Quadriceps CSA was measured at 50% of the segment length with 0.8-cm slices for 3 s. The pulse sequence was performed with a field of view between 400 and 420 mm, time of repetition of 350 ms, eco time from 9 to 11 ms, two signal acquisitions, and a matrix of reconstruction of 256 x 256 mm. The images were then transferred to a workstation (Advantage Workstation 4.3; GE Healthcare) to determine quadriceps CSA. The quadriceps muscle was identified and was delimitated through tracing in triplicates by a specialized researcher, and the mean values were used for further analysis. The coefficient of variation between measurements was 1.89%.

Determination of the Blood Flow Restriction Training Pressure

An 18 cm-wide cuff was placed on the proximal portion of the thigh (inguinal fold region) and inflated until blood pulse absence was observed through auscultation with a vascular Doppler probe (DV-600; Marted Ribeirão Preto, São Paulo, Brazil) placed over the tibial artery. The investigator released the pressure slowly until the first arterial pulse could be detected, which was considered the systolic pressure at the tibial artery. Cuff pressure was set at 50% of the maximum tibial arterial pressure throughout the experimental period. However, subjects repeated this procedure once a week to adjust the training pressure if needed. The average cuff pressure over the training period was of 71 ± 9 mmHg. The cuff was maintained inflated throughout the entire training sessions.

Training protocol

The subjects performed the 45° leg-press exercise (G3-PL70 MATRIX, Indaiatuba, São Paulo, Brazil), two days per week for 12 weeks. The HRT group performed four sets of 10 repetitions with a load corresponding to 70% 1-RM in the first six weeks of training. Load was increased to 80% 1-RM for the remaining weeks. The LRT-BFR group performed a total of four sets, one set of 30 repetitions and three sets of 15 repetitions with a load corresponding to 20% 1-RM in the first six weeks of training. Then, load was increased to 30% 1-RM for the following weeks. A one-minute rest period was granted between sets for both groups. Both training protocols were defined based on the fact that their effectiveness and safety has been individually tested elsewhere (12, 14, 29, 31). The range of motion at the knee joint during the leg-press exercise was of 90°. Subjects were asked to execute the concentric and the eccentric phases of the exercise in two seconds.

Statistical Analysis

Data normality and equality of variance were assessed through the Shapiro–Wilk and Levene tests, respectively. To test between-group values for quadriceps CSA and 1-RM, a one-way ANOVA was performed ($P > 0.05$). A mixed model was performed for each dependent variable assuming group (CG, LRT-BFR, and HRT) and time (pre and post training) as fixed factors and subjects as a random factor. Whenever a significant

F-value was obtained, a Tukey adjustment was performed for multiple comparison purposes. The significance level was set at $P \leq 0.05$. All of the statistical tests were performed using SAS version 9.3 for Windows (SAS Institute Inc., Cary, NC, USA). Although no between-group differences were found at baseline, a visual inspection detected somewhat discrepant leg-press 1-RM values between the experimental groups. Thus, a mixed model assuming groups as a fixed factor, subjects as a random factor, pre-test initial 1RM values as a covariate, and individuals' delta change (%) as dependent variable was used. This analysis was performed to adjust individual delta change values to the covariate (i.e. pre-test values). Then, the estimated mean and standard deviation delta changes (i.e. adjusted by the covariate) from each group were used to calculate effect sizes and not for hypothesis test purposes. Several authors have suggested the use of effect sizes for within- and between-groups comparisons, as they do not give a dichotomic answer (i.e. significant or not significant) and are able to deal with highly variable data (23), Thus, effect size confidence intervals of the differences (ES_{CLdiff}) were calculated using a non-central t distribution to perform within- and between-groups comparisons. Positive and negative confidence intervals [i.e. did not cross zero (0)] were considered as significant. Results are expressed as mean ± standard deviation (SD).

RESULTS

The leg press 1-RM values were significantly increased for the HRT group (PRE: 177 ± 104 Kg, POST: 266 ± 140 Kg; $P < 0.001$) and a trend towards significantly greater values ($P = 0.067$) was observed for the LRT-BFR group (PRE: 273 ± 114 Kg, POST: 316 ± 141 Kg) (Figure 1). The CG showed no differences in the leg press 1-RM values from the pre- to post-training tests (224 ± 81 Kg and 203 ± 84 Kg, respectively; $P = 0.998$). ES_{CLdiff} analysis for the 1-RM data showed an ES of 1.50 (0.78 – 2.41) between HRT and CG. A smaller, though pratically relevant ES (ES: 0.59; CI: 0.03 – 1.22) was found between LRT-BFR and CG. An ES of 0.92 (0.33 – 1.61) was found when comparing HRT and LRT-BFR.

Quadriceps CSA increased significantly over time both in the HRT (PRE: 56.9 ± 14.9 cm^2, POST: 61.1 ± 14.8 cm^2; $P < 0.001$) and LRT-BFR groups (PRE: 67.4 ± 21.5 cm^2, POST: 71.4 ± 22.1 cm^2; $P < 0.001$). No changes in quadriceps CSA were observed in the control group (PRE: 53.6 ± 16 cm^2, POST: 52.7 ± 15.7 cm^2; $P < 0.395$) (Figure 2).

INSERT FIGURE 1 HERE.

Figure 1. Pre- and post-training leg-press 1-RM values for the control group (CG); high–intensity training group (HRT); and low-intensity resistance training with blood flow restriction group (LRT-BFR). *Inset*: Delta percentage (pre to post-training) of each group. Data are presented as mean ± standard deviation. * indicates significant within-group differences (pre- to post-training) ($P \leq 0.05$).

INSERT FIGURE 2 HERE.

Figure 2. Pre- and post-training quadriceps cross sectional area (CSA) for the control group (CG); high-intensity resistance training group (HRT); and low-intensity resistance training with blood flow restriction group (LRT-BFR). *Inset*: Delta percentage (pre- to post-training) of each group. Data are presented as mean ± standard deviation. * indicates significant within-group differences (pre- to post-training) ($P \leq 0.05$)

DISCUSSION

This was the first study to compare the effectiveness between LRT-BFR and HRT on gains in lower-limb muscle function and mass in the elderly. Our main findings were that LRT-BFR and HRT promoted increases in leg-press 1-RM and both methods were similarly effective in inducing increase in quadriceps CSA in a cohort of older individuals. However, relevant differences in strength gains were observed between the two training groups (LRT-BFR: ~17 %; HRT: ~54 %) after the ES analysis.

Decreases in muscle function commonly observed with aging are greatly related to impairments in muscle strength (10, 26). In this respect, strategies aiming to increase muscle strength and preserving functionality in elderly are of interest. HRT has been consistently shown to increase muscle strength in older individuals. For instance, Wallerstein et al. (29) demonstrated that 16 weeks of HRT induced significant improvements in lower-limb maximum dynamic strength. Similarly, Lohne-Seiler et al. (20) reported significant gains in leg-press 1-RM after an 11-week HRT program. In the present study, the HRT group presented significant improvements in leg-press 1-RM (~54 %) whereas the LRT-BFR group had a tendency towards a significant increase ($P = 0.067$; ~17 %). Although the findings of the HRT group are in line with the literature (8, 12), one may speculate that the smaller increase observed in the LRT-BFR group may be somehow related to the lower occlusion pressure (~71 mmHg; 50 % of maximum tibial artery pressure) applied when compared with previous studies in the literature (14, 31). Importantly, the dissonant occlusion pressures are mostly due to the differences in cuff width between studies. In this respect, we opted for a wider cuff (18cm) as it has been previously demonstrated that the wider the cuff, the lower the pressure required to occlude circulation (*e.g.*, an 18-cm wide cuff would require ~140 mmHg to fully occlude blood flow whilst a 4.5-cm cuff requires over 360 mmHg of pressure to induce full blood flow restriction) (6, 30). However, despite the lack of within-group (LRT-BFR) significant differences in the present study, our results are very similar to those of Karabulut et al. (14), who found increases of approximately 20% in leg-press 1-RM after a six-week LRT-BFR utilizing much higher occlusion pressures (~205 mmHg) but using a narrower cuff (5 cm) . In the Karabulut et al. (14) training schemes, they performed three sets (30x15x15) of two leg exercises (leg extension and leg press) with a load corresponding to 20% 1-RM. Thus, they used a training volume (i.e. sets x repetitions) that was the double of the volume used in the present study and achieved similar strength increments, but with only six weeks of training. Furthermore, using a similar training protocol to Karabulut et al. (14), for 12 weeks, Yasuda et al. (31) reported increments of ~33% in leg-press 1-RM. These findings suggest that LRT-BFR may require a higher training volume to produce greater strength increments. Alternatively, the blunted response of LRT-BFR on strength when compared with HRT may be, at least partially, explained by the lesser neural adaptation observed after this training method. In fact, previous studies have demonstrated no changes in the activation level (as assessed by either surface electromyography or twitch-interpolation) of upper-limb muscles after LRT-BFR in healthy young subjects with low loads (20 - 30% 1-RM) (16, 22). Additional studies dedicated to investigate the effects of LRT-BFR on neuromuscular adaptations in the elderly population are necessary to further elucidate this issue.

Maintaining or enhancing skeletal muscle mass is imperative to preserve or rescue impaired functionality with aging (10, 26). Regarding the HRT group, the findings presented herein (*i.e.*, 7.9 % increase in CSA) are within the range reported in the literature. For instance, increases in quadriceps CSA ranging from 6.5 to 10 % have been reported after a short-term HRT (24, 29). Nevertheless, high loads – as those employed in regular HRT – may not be suitable to a part of the elderly population (*e.g.;* frail eldery, novice RT practitioners or even older individuals with joint and/or cardio respiratory impairments) (19, 21, 25). In the present study, the alternative training method (*i.e.*, LRT-BFR) was shown to be just as effective as regular HRT on improving quadriceps CSA (*i.e.;* 6.6 %). Accordingly, Yasuda et al. (31) demonstrated a similar increase (~8.0 %) in muscle CSA, however, it is important to note that the subjects undertook double the exercise volume as compared with the present study, over the same time period (i.e. 12 wks). Although this data is somewhat hard to reconcile, as total volume of exercise has been demonstrated to influence the hypertrophic response to RT (11, 15), it may be suggested that training volume may not be as effective in producing muscle hypertrophy in BFR-RT protocols as when performing regular HRT. This suggestion should be addressed in future studies

In summary, this was the first study to provide data on the direct comparison of LRT-BFR and HRT on a cohort of elderly population, supporting the use of a lower-intensity (and perhaps safer) RT strategy as a surrogate to a high-load RT program. We demonstrated that both training regimes were effective to increase quadriceps CSA and leg-press 1-RM, however, HRT seems to induce greater gains in strength.

PRATICAL APPLICATIONS

Increases in muscle strength and mass are of great important to the maintenance and/or rescue of the independence in elderly. Physical exercising, namely high-intensity resistance training, has been widely recommended as an effective strategy to counteract the losses in functionality and lean mass related with aging. However, an important part of the elderly population present comorbidities that often preclude the usage of high-intensity resistance training, thus, alternative forms of exercising able to induce similar neuromuscular adaptations while preserving feasibility are in need. In this context, low-intensity resistance training (i.e.; 20-30% 1RM) combined with partial blood-flow restriction (50% of the maximum tibial arterial pressure) constitutes an effective alternative approach to high-intensity resistance training in inducing gains in muscle strength and mass in elderly individuals.

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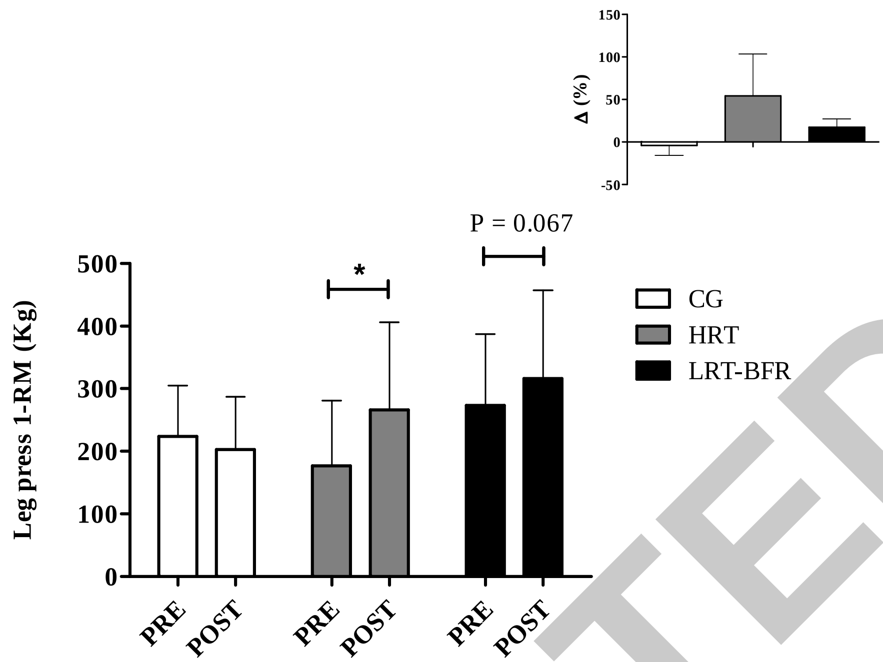

Figure 1. Pre- and post-training leg-press 1-RM values for the control group (CG); high–intensity training group (HRT); and low-intensity resistance training with blood flow restriction group (LRT-BFR). *Inset*: Delta percentage (pre to post-training) of each group. Data are presented as mean ± standard deviation. * indicates significant within-group differences (pre- to post-training) ($P \leq 0.05$).

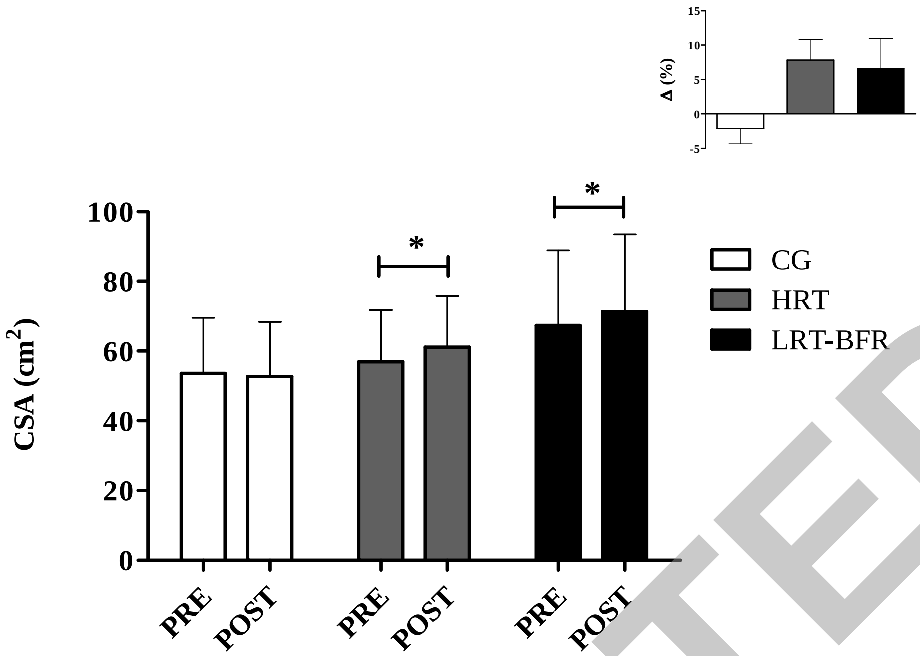

Figure 2. Pre- and post-training quadriceps cross sectional area (CSA) for the control group (CG); high-intensity resistance training group (HRT); and low-intensity resistance training with blood flow restriction group (LRT-BFR). *Inset*: Delta percentage (pre- to post-training) of each group. Data are presented as mean ± standard deviation. * indicates significant within-group differences (pre- to post-training) ($P \leq 0.05$)